

# Python Data Analysis

Python is widely used for data analysis because of libraries such as NumPy, pandas, and Matplotlib.

## NumPy

NumPy provides efficient multidimensional arrays and numerical operations. A NumPy array can store many values and supports vectorized calculations.

## pandas

pandas provides two main data structures: Series and DataFrame. A Series is a one-dimensional labeled array. A DataFrame is a two-dimensional labeled table.

## Reading Data

A CSV file can be loaded using pandas `read_csv`. After loading data, `head` can display the first rows and `info` can show column information and data types.

## Data Cleaning

Common cleaning operations include handling missing values, removing duplicates, changing data types, and renaming columns. Missing values can be checked with `isnull`.

## Filtering and Grouping

Rows can be filtered using conditions. `groupby` is used to group rows and calculate aggregate values such as `sum`, `mean`, `count`, `minimum`, and `maximum`.

## Visualization

Matplotlib is used to create graphs such as line charts, bar charts, scatter plots, and histograms. Visualization helps identify trends and patterns in data.